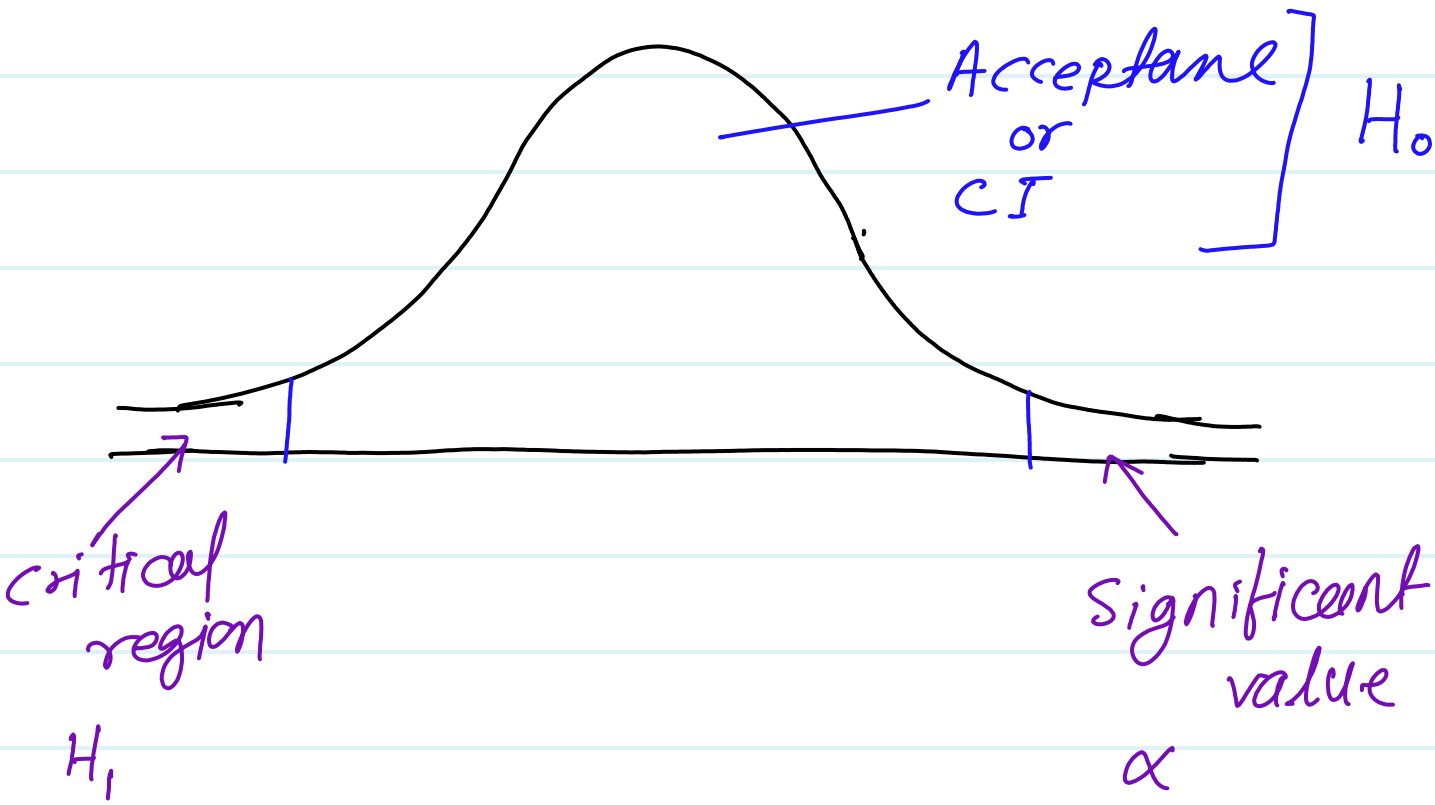
Hypothesis

H_0 – Null hypo

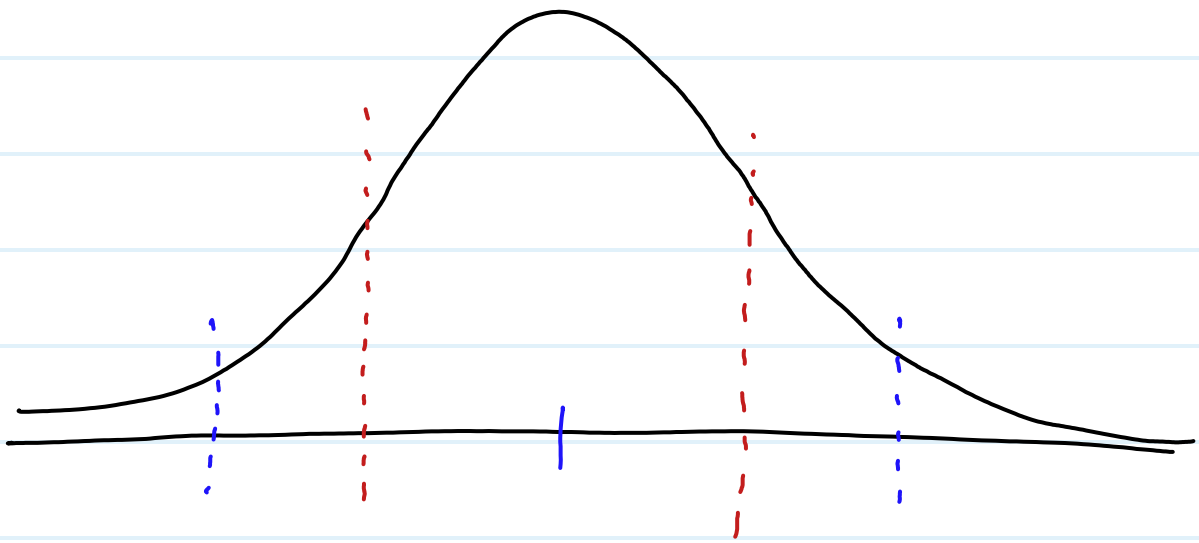
H_1 – Altanet hypo



$\mu = \bar{x}$ – H_0

$\mu \neq \bar{x}$ – H_1

1. one sample
2. Two sample
3. more than two sample
4. one tail test
5. Two tail test



Two tail

$\frac{\alpha}{2}$

100 − 90 CI

$\alpha = 10/2 = 5 - 5$

Z-test F-test Correlation

t-test. chi-square

A/B testing

C.I.

α = value

↑

Domain expert

default 5%

σ
$n > 25$ z-test

s
$n < 25$ t-test

P-value

$\alpha > P$ ✓ — H_1

$\alpha < P$ — H_0

✓ We reject H_0, accept H_1

✓ We fail to reject H_0.

5.49

$n-1 = 4$

$\alpha = 0.05$

$\frac{0.05}{2} = 0.025$

$1 - 0.025 = \boxed{0.975}$

t-stats -1.77 $\quad -1.77 \quad -1.96 \quad \text{———} \quad +1.96$

t-table ± 1.96

$\boxed{p < \alpha}$